

VISHVENA Techno Solutions Pvt. Ltd.

Assignment

Problem Statement:

Imagine that VISHVENA is developing a simple Event Management Web Application.

The application helps an event organizer manage an event and its attendees.

Basic Product Flow

Create Event → Configure Event → Register Attendees → Manage Attendees → Event Check-in → View Reports

The organizer should be able to:

1. Create an event
2. Enter:
 - Event Name
 - Event Date
 - Start Time
 - End Time
 - Location
 - Maximum Attendee Capacity
3. Publish the event
4. Share a registration link
5. Allow attendees to register

6. View registered attendees
7. Check in attendees during the event
8. View basic attendance information

Requirement Analysis

The Event Management Web Application is designed to help an event organizer create, configure, publish, and manage an event. The application should support the complete process from event creation to attendee registration, check-in, and attendance reporting.

Main Users

The application mainly involves two types of users:

- Event Manager/Organizer – Creates and manages the event, publishes it, shares the registration link, manages attendees, performs check-in, and views attendance information.
- Attendee – Uses the registration link to register for the event.

Main Application Flow

Create Event → Configure Event → Publish Event → Register Attendees → Manage Attendees → Check-in → View Report

Key Requirements Identified

The system needs to ensure that:

- All required event information is provided before publishing.
- Event details such as date, time, capacity, and location are properly validated.
- Invalid or conflicting event details are not accepted.
- Duplicate events and duplicate publishing are prevented.
- Only authorized users can perform organizer-level actions.
- Attendees can register through the event's registration link.
- Registration should follow the event's maximum capacity.
- Registered attendees can be managed and checked in during the event.
- Attendance information should be accurately maintained and reflected in reports.

- The system should maintain consistency between attendee check-in information and attendance reports.

Functional Requirements

1. The system should not allow an event to be published without a start time and an end time.
2. The system should not allow an event to be published without specifying the maximum attendee capacity.
3. The system should allow an event to be published only when the event date has been entered.
4. The system should not allow an event to be published if another event with the same details already exists.
5. The system should not allow an event to be published when the start time and end time are the same.
6. The system should allow an event to be published only when the start time and end time are entered in a valid 12-hour or 24-hour format.
7. Only the event manager/organizer should be allowed to publish an event. Other members should not have permission to publish the event.
8. The system should allow an event to be published only once and should prevent the same event from being published multiple times with the same details, such as event date, start time, end time, and location.
9. The system should allow the registration link to be shared only by the person who created the event.
10. The system should allow only the person who created the event to view attendees, check in attendees, and view attendance information.

Business Rules

1. Two events cannot be published with the same event date, start time, end time, and location.
2. Only the event manager/organizer is authorized to publish an event.
3. An event can be published only once.
4. Only the event creator is authorized to share the registration link.

5. Only the event creator is authorized to access attendee and attendance information.
6. An attendee can register for the same event only once using the same email address.
7. An attendee must be registered for the event before they can be checked in.

Test scenarios

Test Scenario ID	Module	Test Scenario	Type	Expected Outcome
TS-001	Event Creation	Create an event by entering all required details correctly.	Positive	Event should be created successfully.
TS-002	Event Validation	Try to publish an event without entering the event date.	Negative	System should prevent publishing and display a validation message.
TS-003	Event Validation	Try to publish an event without entering the start time and end time.	Negative	System should prevent publishing and indicate that the time details are required.
TS-004	Event Validation	Try to publish an event without specifying the maximum attendee capacity.	Negative	System should prevent publishing and request the maximum capacity.
TS-005	Event Validation	Enter the same start time and end time for an event.	Boundary / Negative	System should prevent publishing because the

				event duration cannot be zero.
TS-006	Event Validation	Enter start and end times using a valid 12-hour format.	Positive	System should accept the time format.
TS-007	Event Validation	Enter start and end times using a valid 24-hour format.	Positive	System should accept the time format.
TS-008	Event Validation	Enter an invalid time format while creating an event.	Negative	System should reject the invalid time format and display a validation message.
TS-009	Event Publishing	Publish a valid event as the event manager/organizer.	Positive	Event should be published successfully.
TS-010	Event Publishing	Attempt to publish an event as a user who is not authorized to publish events.	Negative	System should deny the publishing action.
TS-011	Event Publishing	Attempt to publish the same event more than once.	Boundary / Negative	System should prevent duplicate publishing of the same event.
TS-012	Event Publishing	Try to publish another event with the same date, start time,	Negative	System should prevent the duplicate event

		end time, and location as an existing event.		from being published.
TS-013	Attendee Registration	Register an attendee using valid name and email details through the registration link.	Positive	Registration should be completed successfully and the attendee should appear in the attendee list.
TS-014	Attendee Registration	Attempt to register the same attendee for the same event more than once using the same email address.	Negative	System should reject the duplicate registration.
TS-015	Capacity Handling	Register attendees until the event reaches its maximum capacity.	Boundary	Registrations should be accepted up to the exact capacity limit.
TS-016	Capacity Handling	Attempt to register another attendee after the maximum capacity has been reached.	Negative	System should prevent registration because the event is full.
TS-017	Attendee Management	Search for and view a registered attendee using their name or email address.	Positive	The correct attendee information should be displayed.

TS-018	Attendee Management	Attempt to view attendee information as a user who is not the event creator.	Negative	System should deny access to attendee information.
TS-019	Event Check-in	Check in a registered attendee during the event.	Positive	Attendee status should change to Checked In.
TS-020	Event Check-in	Attempt to check in an attendee who is not registered for the event.	Negative	System should prevent the check-in.
TS-021	Event Check-in	Attempt to check in the same attendee more than once.	Boundary / Negative	System should prevent duplicate check-in and retain the existing check-in status.
TS-022	Event Check-in	Attempt to check in an attendee as a user who is not authorized to manage the event.	Negative	System should deny the check-in action.
TS-023	Reports	View the attendance report after several attendees have registered and checked in.	Positive	Report should correctly display registered, checked-in, and remaining attendees.
TS-024	Reports	Compare the attendance	Positive	Report information

		report with the actual attendee check-in records.		should accurately match the attendee check-in status.
TS-025	Reports	Attempt to view attendance information as a user who is not the event creator.	Negative	System should deny access to attendance information.

Detailed test cases

Test Case ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Priority
TC-001	Create an event by entering all required details correctly.	User is logged in as an event organizer and is on the event creation page.	1. Enter the event name. 2. Select a valid future event date. 3. Enter the start time. 4. Enter the end time. 5. Enter the location. 6. Enter the maximum attendee capacity. 7. Click Create/Publish.	Event Name: Tech Workshop Date: 15-Nov-2026 Start: 10:00 AM End: 1:00 PM Location: Bengaluru Capacity: 100	The event should be created and published successfully with the entered details.	High
TC-002	Prevent publishing an event when the event date is missing.	User is logged in as an event organizer and is on the event creation page.	1. Enter a valid event name. 2. Leave the event date blank. 3. Enter valid start and end times. 4. Enter location and capacity. 5. Click Publish.	Event Date: Blank Start: 10:00 AM End: 1:00 PM Capacity: 100	The system should prevent publishing and display a message indicating that the event date is required.	High
TC-003	Prevent publishing an event when the start time	User is logged in as an event organizer and has entered all	1. Enter a valid event date. 2. Enter the same start and end time.	Start Time: 10:00 AM End Time: 10:00 AM	The system should reject the event and display a validation	High

	and end time are the same.	other required event details.	3. Enter the remaining required details. 4. Click Publish.		message indicating that the start and end times cannot be the same.	
TC-004	Prevent unauthorized users from publishing an event.	An event has been created with valid details. User is logged in without organizer/event-manager permissions.	1. Open the event. 2. Attempt to access the Publish option. 3. Try to publish the event.	User Role: Unauthorized User	The system should deny the publishing action and prevent the event from being published.	High
TC-005	Prevent duplicate registration using the same email address.	A valid event has been published and its registration link is available. An attendee has already registered for the event.	1. Open the event registration link. 2. Enter the same attendee name and email address used for the previous registration. 3. Submit the registration.	Name: Rahul Kumar Email: rahul@example.com	The system should reject the duplicate registration and display an appropriate message.	High
TC-006	Prevent registration after the maximum capacity is reached.	A published event has a maximum capacity of 100. 100 attendees have already registered successfully.	1. Open the event registration link as a new attendee. 2. Enter valid attendee details. 3. Submit the registration.	Capacity: 100 Existing Registrations: 100 New Attendee: Ananya Sharma Email: ananya@example.com	The system should reject the registration and indicate that the event has reached maximum capacity.	Critical
TC-007	View registered attendee information.	A published event has registered attendees. User is logged in as the event creator.	1. Open the event management page. 2. Select Attendees. 3. Search for an attendee by name or email.	Name: Rahul Kumar Email: rahul@example.com	The system should display the correct attendee information.	Medium
TC-008	Check in a registered attendee.	The event is active and the attendee has already registered. User is logged in as the event creator.	1. Open the event's attendee list. 2. Search for the registered attendee. 3. Select the attendee. 4. Click Check In.	Attendee: Rahul Kumar Email: rahul@example.com Status: Registered	The attendee's status should change to Checked In successfully.	High

TC-009	Verify that attendance report reflects the attendee's check-in status.	A registered attendee has been successfully checked in. User is logged in as the event creator.	1. Check in the registered attendee. 2. Open the attendance report. 3. Locate the attendee in the report. 4. Compare the report status with the attendee's check-in status.	Attendee: Rahul Check-in Status: Checked In	The attendance report should show the attendee as Checked In, matching the actual check-in record.	High
TC-010	Prevent an unauthorized user from accessing attendance information.	The event has registered attendees. User is logged in without event-creator/organizer permissions.	1. Open the event. 2. Attempt to access attendee or attendance information. 3. Attempt to view the attendance report.	User Role: Unauthorized User	The system should deny access to attendee and attendance information.	High

Bug Report

Bug ID	Title	Steps to Reproduce	Expected Result	Actual Result	Severity	Priority
BUG-001	Event can be created with an end time earlier than the start time	1. Open the event creation page. 2. Enter all required event details. 3. Enter a start time of 5:00 PM. 4. Enter an end time of 3:00 PM. 5. Click Create/Publish.	The system should reject the event and display a validation message because the end time is earlier than the start time.	The system allows the event to be created successfully with an invalid time range.	High	High
BUG-002	Invalid email address is accepted	1. Open the event registration link.	The system should reject the	The registration is	High	High

	during registration	2. Enter a valid attendee name. 3. Enter an invalid email address such as abc@. 4. Submit the registration.	registration and display an appropriate email validation message.	completed successfully even though the email address is invalid.		
BUG-003	System allows registration beyond maximum capacity	1. Set the event capacity to 100. 2. Register 100 attendees successfully. 3. Attempt to register a 101st attendee.	The system should reject the 101st registration and indicate that the event has reached maximum capacity.	The 101st attendee is also registered successfully.	High	High
BUG-004	Check-in status is not reflected correctly in attendance report	1. Register an attendee. 2. Check in the attendee. 3. Verify that the attendee is shown as Checked In. 4. Open the attendance report.	The attendance report should show the attendee as Checked In.	The attendee is shown as Not Checked In in the attendance report.	High	High
BUG-005	Double-clicking Publish creates duplicate events	1. Enter valid event details. 2. Click the Publish button twice. 3. Open the event list.	Only one event should be created and published.	Two identical events are created.	High	High

Severity and Priority Justification

BUG-001 – High Severity / High Priority

The system allows an event to be created with an invalid time range, which can cause incorrect event scheduling and confusion for organizers and attendees. Therefore, the bug is considered High Severity. It should be fixed quickly because event creation is a core functionality, so it has High Priority.

BUG-002 – High Severity / High Priority

The system accepts an invalid email address and completes the registration. This can result in incorrect attendee information and may prevent important event communication from reaching the attendee. Therefore, it is High Severity and should be fixed with High Priority.

BUG-003 – Critical Severity / Critical Priority

The system allows registration beyond the maximum event capacity. This directly violates the capacity rule and can result in more attendees than the venue or event can accommodate. It can create serious operational and safety issues. Therefore, it is Critical Severity and requires Critical Priority.

BUG-004 – High Severity / High Priority

The check-in status and attendance report show different information. This makes the attendance report unreliable and can result in incorrect attendance records. Therefore, it is High Severity and should be fixed with High Priority.

BUG-005 – High Severity / High Priority

Clicking the Publish button twice creates two identical events. This can lead to duplicate event records, duplicate registrations, and incorrect reports. Since publishing is an important event-management function, the bug is High Severity and should be fixed with High Priority.

Release decision

I would not recommend releasing the product tomorrow in its current state. The product has 2 Critical, 3 High-priority, 8 Medium-priority, and 12 Low-priority issues. The Critical issues could have a major impact on core functionality and should be resolved and tested before release.

Additional Information Required

- What functionality is affected by the Critical and High-priority bugs?
- Are the Critical bugs reproducible?

- Are there any workarounds?
- Have the fixes already completed passed regression testing?
- Has UAT been completed and approved?

Issues to Investigate First

1. 2 Critical bugs – fix and retest immediately.
2. 3 High-priority bugs – assess their impact and fix important issues before release.
3. Medium and Low issues – determine which can safely be deferred.

Risks to Communicate

- Critical bugs may cause major failures in core features.
- Incorrect registration, check-in, or attendance information could affect organizers and attendees.
- Last-minute fixes may introduce new defects if proper regression testing is not performed.

Release Conditions

I would consider release only if:

- Critical bugs are fixed and successfully retested.
- Important High-priority bugs are resolved or have an approved workaround.
- Regression testing is completed successfully.
- Major workflows such as event creation, registration, check-in, and reporting work correctly.
- UAT is approved and the remaining risks are accepted by the stakeholders.

Final Decision

NO-GO for release at this stage. I would recommend delaying the release until the Critical issues are fixed and verified. After successful regression testing and stakeholder approval, the release decision can be reconsidered.

Risk / blocker / dependency analysis

Project Risks

1. Requirement changes after development has started

- Possible Impact: May cause rework, additional development time, and project delays.
- What would you do: Discuss the change with the Project Manager, assess its impact, and prioritize the change based on project needs.

2. Backend development delay

- Possible Impact: API development and frontend integration may be delayed.
- What would you do: Coordinate with the backend team and Project Manager, identify the critical pending tasks, and revise priorities if required.

3. High number of defects during QA

- Possible Impact: Bug fixing and regression testing may take more time and delay the release.
- What would you do: Prioritize Critical and High-severity defects first and closely monitor the bug-fixing progress.

4. API development delay

- Possible Impact: Frontend integration and testing may be affected.
- What would you do: Coordinate with the API and frontend teams and prioritize the APIs required for critical features.

5. Major issues found during UAT

- Possible Impact: The production release may have to be postponed.
- What would you do: Assign owners for the issues, prioritize them, and update the release plan based on the resolution status.

Project Blockers

1. Test environment is unavailable

- Possible Impact: QA team cannot start or continue testing.

- What would you do: Inform the Project Manager and coordinate with the responsible team to restore the environment as quickly as possible.

2. Required API is unavailable

- Possible Impact: Frontend integration and related testing may be blocked.
- What would you do: Coordinate with the API team and use mock APIs if possible so development can continue.

3. Critical bug blocks a major workflow

- Possible Impact: QA or UAT cannot continue with important test scenarios.
- What would you do: Escalate the issue, assign it to the development team, and prioritize the fix.

Project Dependencies

1. Frontend depends on UI/UX completion

- Possible Impact: UI changes or delays can affect frontend development.
- What would you do: Confirm the designs early and communicate any changes immediately to the frontend team.

2. Frontend integration depends on API development

- Possible Impact: Frontend features may not work correctly until the required APIs are available.
- What would you do: Track API progress and prioritize the APIs required for frontend integration.

3. UAT depends on successful QA and regression testing

- Possible Impact: UAT may be delayed if major defects are still unresolved.

- What would you do: Complete Critical and High-priority testing and ensure a stable build is available before starting UAT.

Project Status Report

Overall Status

The project is currently **AMBER / At Risk**. UI/UX is completed, but development is behind schedule by 3 days. API development is only 60% complete, which is affecting two developers who are waiting for API completion. Since UAT is planned for next week, the remaining development and testing activities need to be completed on priority.

Completed

- UI/UX design is **100% completed**.

In Progress

- Frontend development – **80% completed**
- Backend development – **70% completed**
- API development – **60% completed**

Delayed

- Development team has reported a **3-day delay**.
- QA has not started yet and may be affected by the development and API delay.

Blockers

- API completion is currently blocking **two developers** who are waiting for the required APIs.
- QA cannot start properly until the required development and API work is completed.

Risks

- Further API delays may affect frontend integration.
- Less time may be available for QA and regression testing.
- Late discovery of defects could affect the planned UAT next week.
- The overall project release may be delayed if the current delay continues.

Dependencies

- Frontend integration depends on **API completion**.
- QA depends on the completion of major **frontend, backend, and API development**.
- Bug fixing depends on QA identifying defects.
- Regression testing depends on successful bug fixing.
- UAT depends on a stable build after QA and regression testing.

Next Actions

1. Prioritize the remaining critical API development.
2. Coordinate with the API and development teams to remove the current dependency.
3. Set a firm date for completing the required APIs.
4. Prepare QA test data and test cases while development is being completed.
5. Start QA as soon as a stable build is available.
6. Monitor progress daily and immediately escalate any issue that could affect UAT next week.

Communication exercise

Subject: Follow-up on Pending Task

Dear Sir,

I wanted to check the status of the Event Registration Module, which was due yesterday. Could you please let me know how much of the task is completed?

Please let me know if you are facing any blockers. If possible could you tell me when you expect to complete the task?

Thank you.